

4EB00 Special Topic Thermodynamics and Combustion

Cycle analysis of a jet-engine
Lecture 2

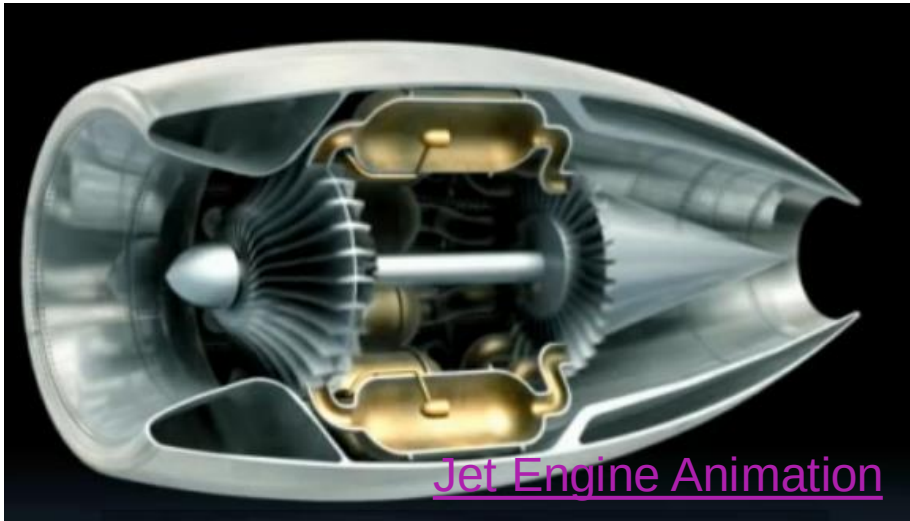
Bart Somers (Combustion Technology)



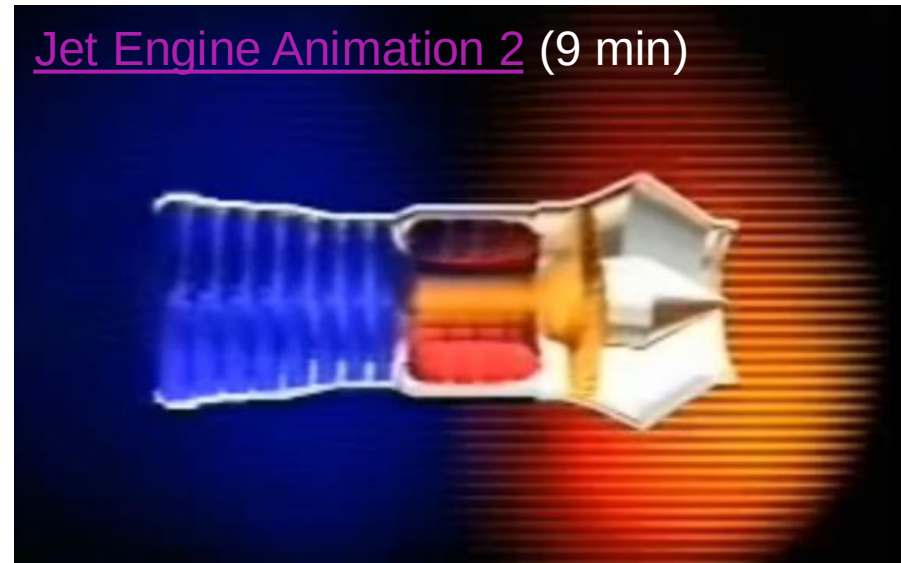
Technische Universiteit
Eindhoven
University of Technology

Where innovation starts

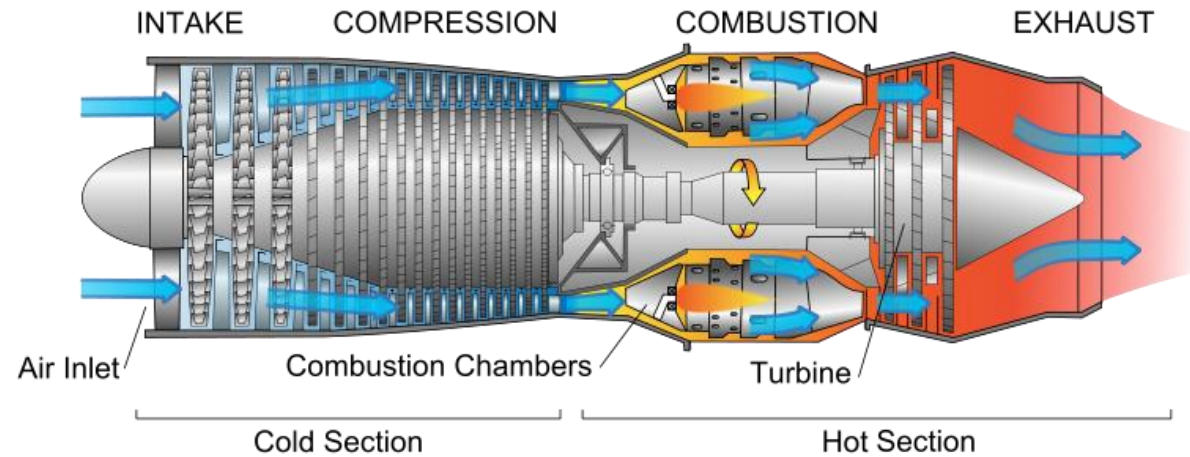
An animation



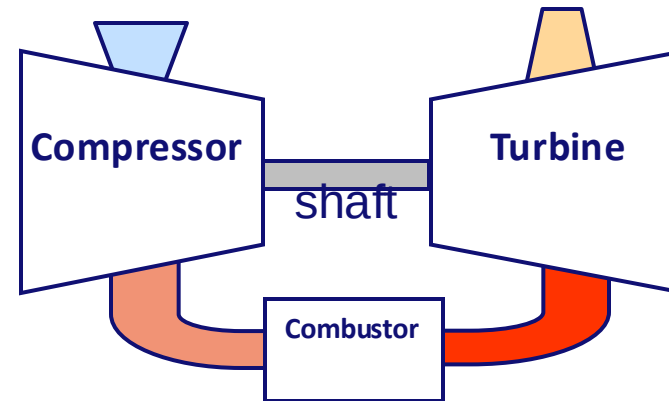
Also see [youtube link to Turbo fan engines](#)
For a nice explanation of
modern jet engines: turbo fan engines



The Jet Engine



- **Control volume models**

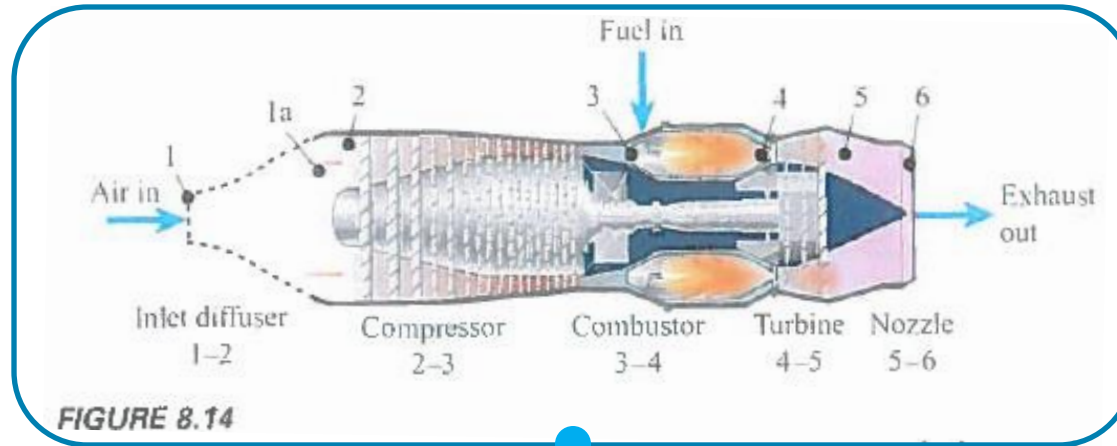


Contents

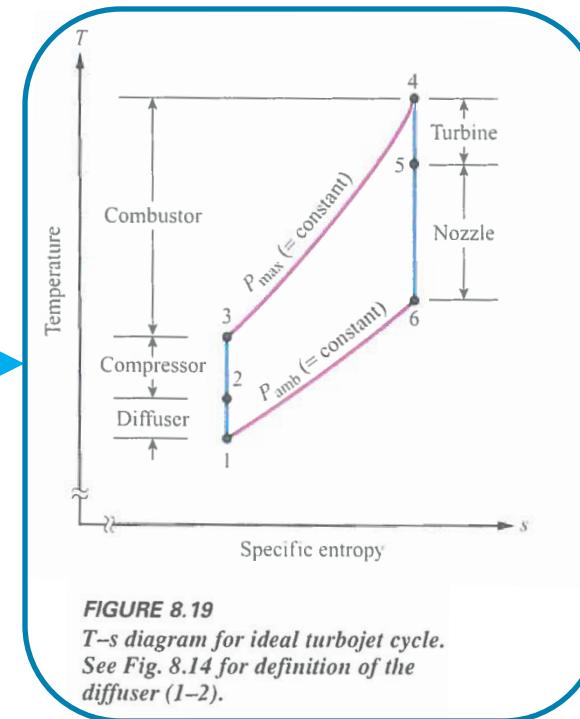
- Ideal Gas mixtures
 - Thermodynamic Properties of a mixture (2.9)
- Cycle analysis of a jet engine using thermodynamic tables
 - Combustor
 - Diffusor
 - Etc..
 - **WARNING:** in your model there should be **no** need to use Poisson relations ()

$$pV^\gamma = \text{constant}$$

Cycle analysis of a jet engine using thermodynamic tables



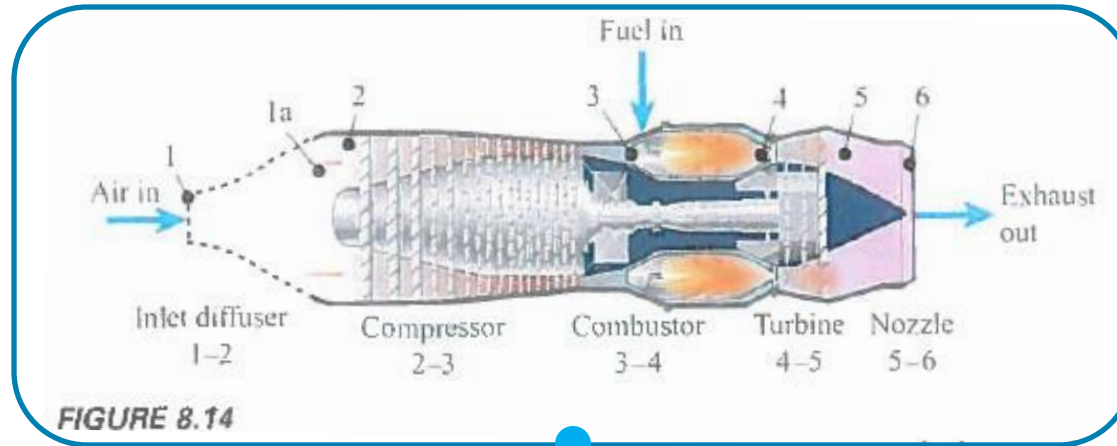
Using Nasa tables



Contents

- **Ideal Gas mixtures**
 - Thermodynamic Properties of a mixture (2.9)
- **Cycle analysis of a jet engine using thermodynamic tables**
 - Combustor
 - Diffusor
 - Etc..

Cycle analysis of a jet engine using thermodynamic tables



Using Nasa functions

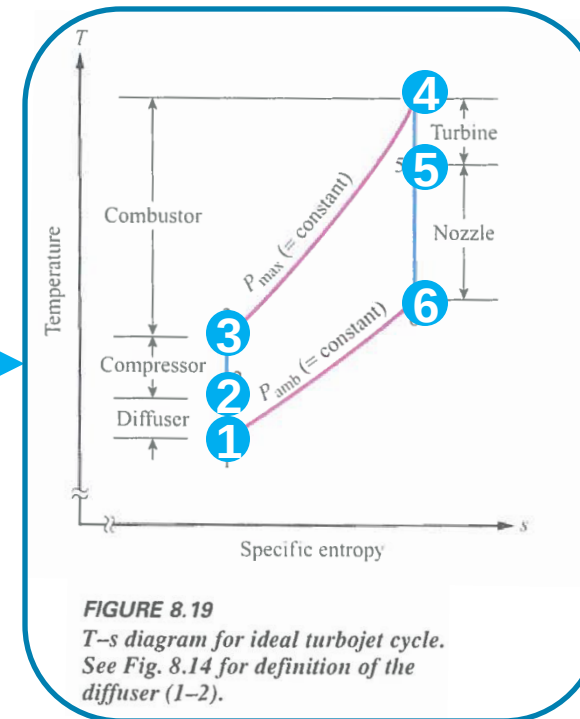
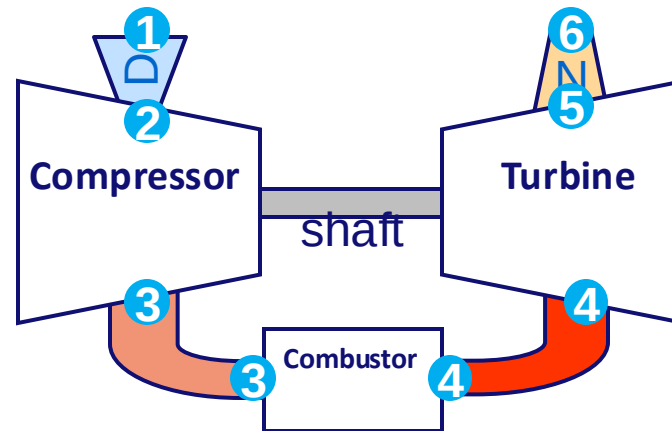
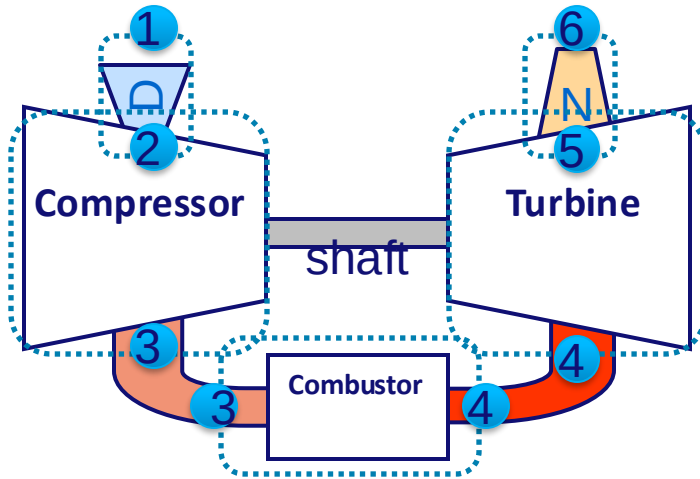


FIGURE 8.19
 $T-s$ diagram for ideal turbojet cycle.
See Fig. 8.14 for definition of the diffuser (1-2).

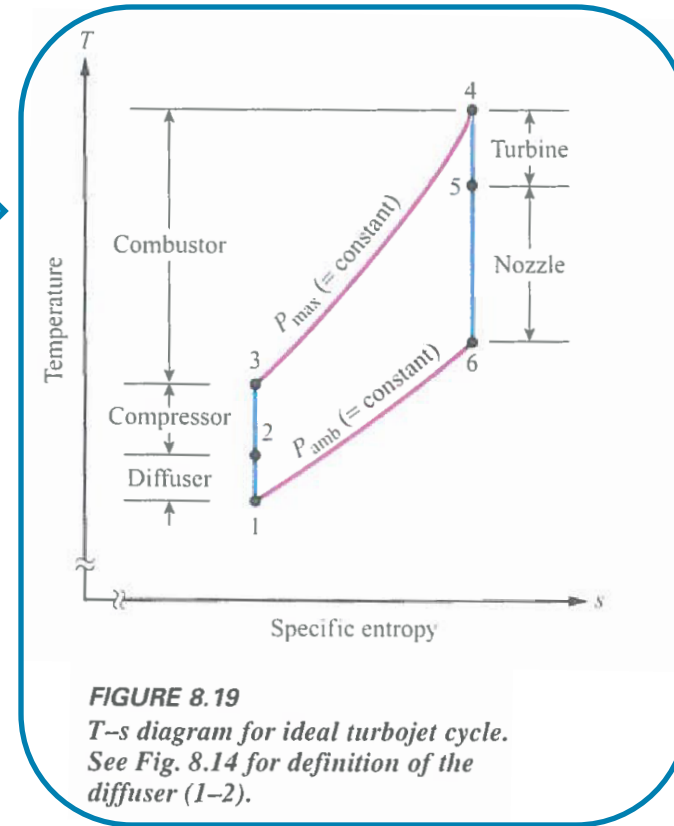
Cycle analysis of a jet engine using thermodynamic tables



Apply thermodynamics on every control volume,
see table 7.1 in book

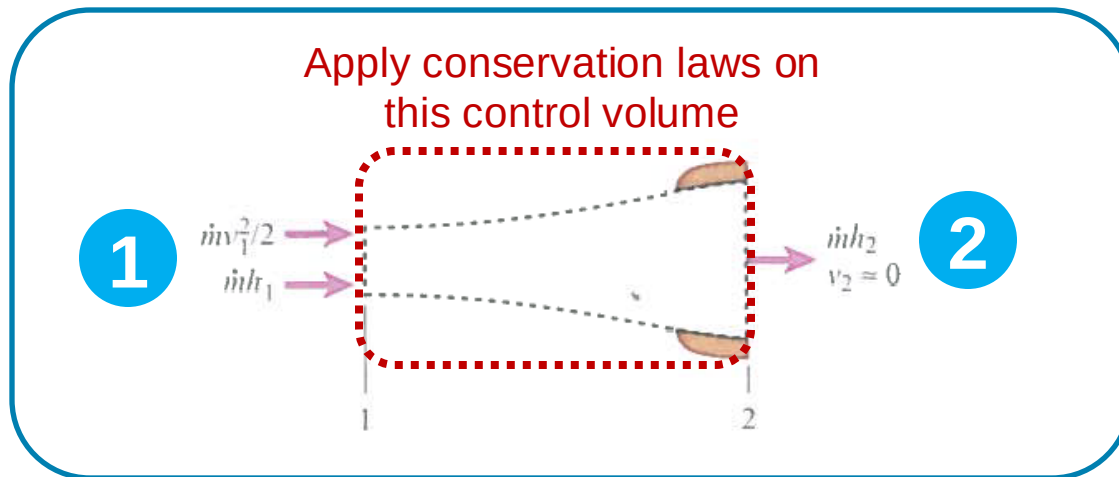
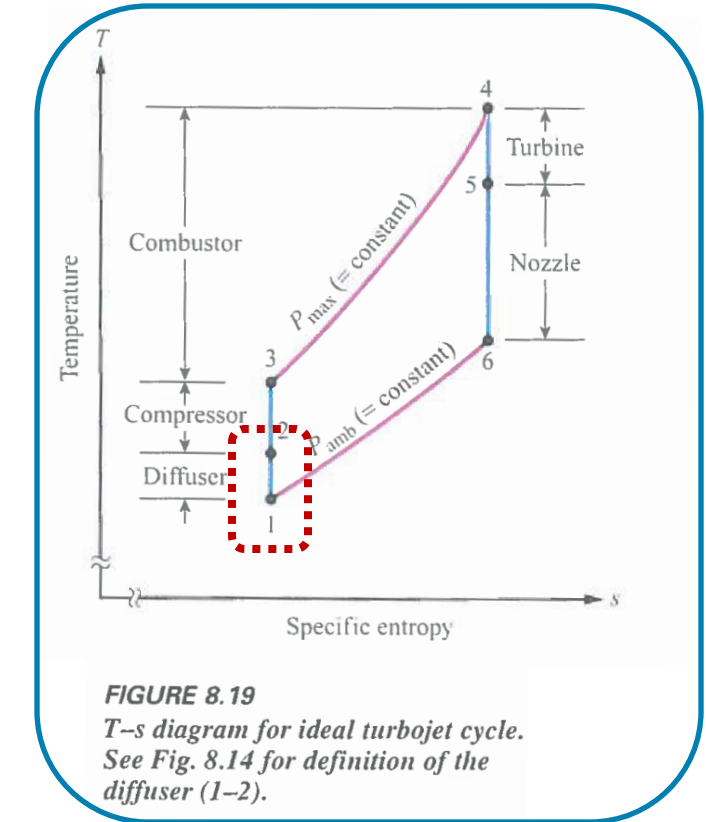
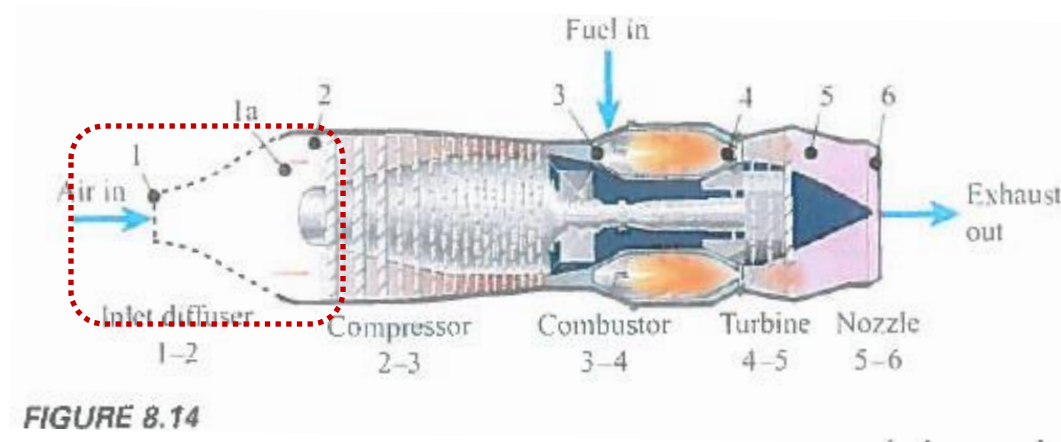
Given is:

- $P_1 = P_{amb}$, $T_1 = T_{amb}$
- $v_1 = \text{flight speed}$
- Compressor ratio P_3/P_2 is known
- T_4 is restricted (must be kept below certain value to prevent turbine blade failure)
- $P_6 = P_{amb}$
- Isentropic efficiency of each component is given.



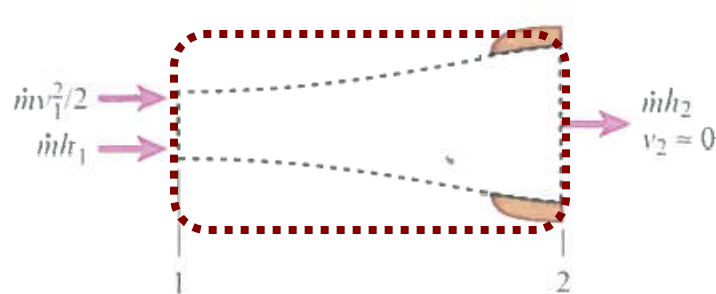
Note: in Turns T_4 is given,
in your model it
should be computed

Cycle analysis of a jet engine using thermodynamic tables



Cycle analysis of a jet engine using thermodynamic tables

Diffuser 1-2:

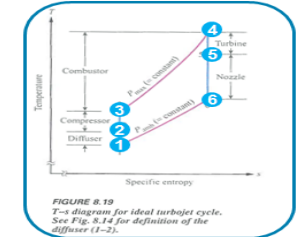
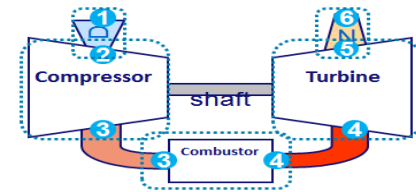


1. Mass conservation
2. Energy conservation
3. Isentropic

$$\begin{aligned}
 1+2 &: h_2(T_2) = h_1(T_1) + \frac{1}{2} v_1^2 \quad \text{given} \\
 3 &: s_2(T_2, P_2) - s_1(T_1, P_1) = 0 \Rightarrow \int_{T_1}^{T_2} \frac{c_p}{T'} dT' - \int_{P_1}^{P_2} \frac{R_g}{P'} dP' = 0 \rightarrow P_2
 \end{aligned}$$

unknown

Cycle analysis of a jet engine using thermodynamic tables



Apply thermodynamics on every control volume, see table 7.1 in book

Given is:

- $P_1 = P_{amb}$
- $v_1 = \text{flight speed}$
- Compressor ratio P_3/P_2 is known
- T_4 is restricted (must be kept below certain value to prevent turbine blade failure)
- $P_6 = P_{amb}$
- Isentropic efficiency of each component is given.

/ Mechanical Engineering

TU/e Technische Universiteit Eindhoven University of Technology

Q1 2015 PAGE 13

Cycle analysis of a jet engine using thermodynamic tables

Intermezzo: Nasa polynomials functions (general folder)

Property	Function Name	Unit
c_p	CpNasa	[J/kg/K]
c_v	CvNasa	[J/kg/K]
h	HNasa	[J/kg]
u	UNasa	[J/kg]
s_T	SNasa	[J/kg]

Temperature part of specific entropy

$$\begin{array}{ll}
 1+2 & : \quad h_2 = h_1 + \frac{1}{2} v_1^2 \\
 3 & : \quad \int_{T_1}^{T_2} \frac{c_p}{T'} dT' - \int_{P_1}^{P_2} \frac{R_g}{P'} dP' = 0
 \end{array}$$

$$p \cdot T^{\frac{\gamma}{1-\gamma}} = \text{const}$$

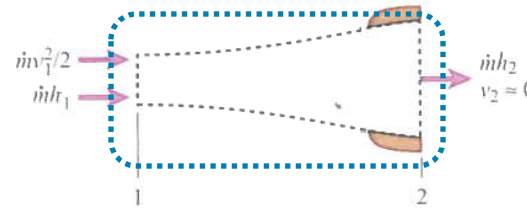
Note: equation 3 replaces the **Poisson relation** used in Turns. In fact that specific Poisson relation simply follows from equation 3 if you would take C_p constant!

Cycle analysis of a jet engine using thermodynamic tables

Diffuser 1-2:

Code snippet

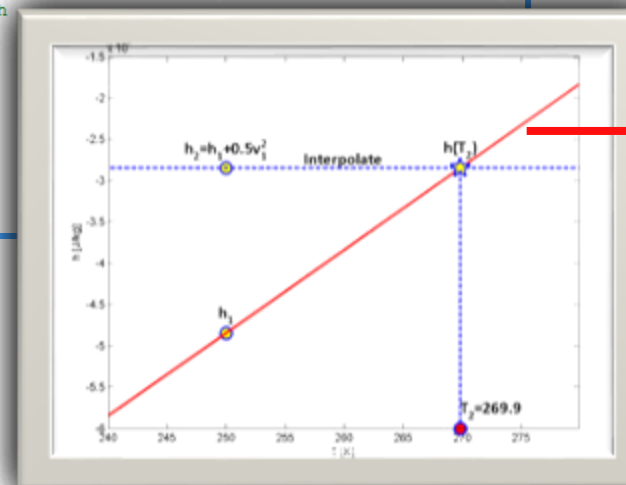
```
%% Range of enthalpies/thermal part of entropy of species
TR = [200:50:3000];
for i=1:NSp
    hia(:,i) = HNasa(TR,SpS(i));
    sia(:,i) = SNasa(TR,SpS(i));
end
hair_a = Yair*hia'; % enthalpy of air for range of T
sair_a = Yair*sia'; % thermal part of entropy of air for range of T
%% [1-2] Diffusor
sPart = 'Diffusor';
T1 = Tamb;
P1 = Pamb;
Rg = Runiv/MAir;
for i=1:NSp
    hi(i) = HNasa(T1,SpS(i));
end
h1 = Yair*hi';
h2 = h1+0.5*v1^2;
T2 = interp1(hair_a,TR,h2); % Not exactly correct but nearly. Why??? can also do a search
for i=1:NSp
    hi2(i) = HNasa(T2,SpS(i));
    si1(i) = SNasa(T1,SpS(i));
    si2(i) = SNasa(T2,SpS(i));
end
h2check = Yair*hi2';
s1thermal = Yair*si1';
s2thermal = Yair*si2';
lnPr = (s2thermal-s1thermal)/Rg;
Pr = exp(lnPr);
P2 = P1*Pr;
```



$$h_2 = h_1 + \frac{1}{2} v_1^2$$

$$s_2 - s_1 = \int_{T_1}^{T_2} \frac{c_p}{T'} dT' - \int_{P_1}^{P_2} \frac{R_g}{P'} dP'$$

$$SNasa(T, SpS(i)) = \int_{T_{ref}}^T \frac{c_{p,i}}{T'} dT'$$



Cycle analysis of a jet engine using thermodynamic tables

Assignment

Complete the Template for all control volumes as they are specified in the book but use the Thermodynamical Database

Different groups have different conditions so do not copy.

Will check some codes on copycatting. Details matter.

Group settings are provided in a zip-file on Canvas

▼ Lecture material - Special Topic 1	✓	+	⋮
4EB00 Special Topic Jet Engine Info 2020.pdf	✓		⋮
4EB00 Lecture Cycle analysis 2020.pdf	✓		⋮
4EB00 Lecture Ideal gas mixtures 2020.pdf	✓		⋮
4EB00 Special Topic Jet Engine Report Template 2020.docx	✓		⋮
JetEngineMatlab.zip	✓		⋮
GroupSettings.zip	✓		⋮

Template 4EB00
Special Topic: The Jet Engine

Groep ?

	Naam	Studentnr
1		
2		

Assignment 1 The adiabatic flame temperature

Code snippet

Jet Engine (4EB00) 2015



Cycle analysis of a jet engine using thermodynamic tables

Assignment

GroupSettings.zip (Pick your number!)

Groep001.txt

Groep002.txt

Groep003.txt

Groep004.txt

Groep005.txt

Groep006.txt

Groep007.txt

Groep008.txt

Groep009.txt

Groep010.txt

Groep011.txt

Groep012.txt

Groep013.txt

Groep014.txt

Groep015.txt

Groep016.txt

Groep017.txt

Groep 11

Data

Fuel : CH4

Tamb [K] : 250.00

P3overP2 [-] : 8.00

Pamb [Pa] : 55000.00

mfurate [kg/s] : 0.68

AF [-] : 102.78

v1 [m/s] : 200.00

'Groep' is Dutch for 'Group'

Cycle analysis of a jet engine using thermodynamic tables

Assignment Deadlines

Hand-in pdf report and zip-file with scripts (such that runs)

07-10 2025: Regular submission

14-10 2025: Late submission $\text{Grade} = \min(\text{Grade}, 8)$

Example

report grade 9 → final grade 8

report grade 7 → final grade 7